

4. (1.75 points)

Given the following information for an insurer's book of business as of December 31, 2015:

Accident <u>Year</u>	On-Level Premium <u>(\$000)</u>	Paid Losses <u>(\$000)</u>
2012	800	480
2013	1,000	530
2014	1,500	640
2015	1,250	290

- The expected loss payment pattern for the insurance company was approximated by the following function, where G is the cumulative proportion of ultimate losses paid and x represents the average age (in months) since accident occurrence:

$$G(x) = \frac{x^{1.1}}{x^{1.1} + 8.0^{1.1}}$$

- The expected loss ratio (ELR) is 62.5% for this book.

a. (0.75 point)

Use the Cape Cod method to calculate the expected unpaid losses for accident year 2013.

b. (1 point)

Evaluate the appropriateness of using the Cape Cod method with a constant ELR for this book of business.

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QUESTION 4

TOTAL POINT VALUE: 1.75

LEARNING OBJECTIVE: A2: Estimate parameters and unpaid claims using claims development models related to loss reserving methods such as chain ladder, Cape Cod, chain ladder plus calendar-year effects, and Bornhuetter-Ferguson.

SAMPLE ANSWERS

Part a: 0.75 point

Sample Answer 1

Average Age (x) = 36 – 6 = 30

G(x) = 0.811

Premium × ELR for 2013 = \$1,000,000 × 62.5% = \$625,000

Unpaid Losses = \$625,000 (1 - 0.811) = \$118,125

Sample Answer 2

Assume no truncation needed.

2013 -> 30 months

G(30) = 30^{1.1}/(30^{1.1}+8^{1.1}) = 81.06%

2013 expected unpaid = 1000 × 62.5% × (1-81.06%) = 118,375

Part b: 1 point

Sample Answer 1

AY	Used Up Premium	Loss	Loss / Unused Prem
2012	688.8	480	0.697
2013	811	530	0.654
2014	1063.5	640	0.602
2015	526.875	290	0.550

There is an obvious downward trend in the loss ratios emerged to date by AY (more recent AY show better loss performance). This implies that using an all year combined ratio may not be appropriate as it will overstate reserves for recent years and understate for older AY.

Sample Answer 2

Graph the expected paid total minus the paid to date vs time and expect to see consistency if they are a constant ELR.

AY	Avg Age	G(x)	Expected Paid	Paid	Difference
2012	42	0.861	430.5	480	-49.5
2013	30	0.811	506.875	530	-23.125
2014	18	0.709	664.69	640	24.69
2015	6	0.422	329.688	290	39.688

Expected Paid = On-level Prem × ELR × G(x)

Instead they are increasing with time so it is not appropriate.

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EXAMINER'S REPORT

Part a

Candidates were expected to know how to calculate the expected unpaid losses for a single accident year using the Cape Cod method. To receive full credit, a candidate needed to calculate the correct average age of losses for 2013, then use that in the $G(x)$ formula given, then to use that result in the Cape Cod formula to determine the unpaid losses.

Common errors included using the average loss ratio for all four years rather than the loss ratio given and calculating a total for four years rather than just the 2013 year asked for.

Many candidates chose to use the pattern truncation discussed in the reading. This was not necessary, and it neither earned nor lost credit.

Part b

Candidates were expected to know whether or not the Cape Cod method was appropriate when loss ratios were declining. To receive full credit, candidates had to calculate the *a priori* loss ratio (which includes a calculation of the used up premium). Candidates had to deduce that the declining pattern in the loss ratios indicated a bias that made the method inappropriate. The *a priori* loss ratio could be calculated more than one way for full credit.

Common errors included not calculating any loss ratios and instead attempting to deduce the answer by looking at the given premiums (this provides enough evidence to trigger a look into the appropriateness of the Cape Cod method but not enough to make the determination), calculating the loss ratio incorrectly, and not stating a position on whether or not the method was appropriate.